

SDS 8102: Advanced Mathematical Statistics

Solution Key: Laws of Large Numbers & Empirical Processes

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Mohamed bin Zayed University of Artificial Intelligence

1 A Heavy-Tailed Moving-Average Process

Let $m = \mathbb{E}[Y_t]$.

1. The Pareto density is

$$f(y) = \alpha y^{-\alpha-1} \mathbf{1}\{y \geq 1\}.$$

Thus

$$\mathbb{E}[Y_t] = \alpha \int_1^\infty y^{-\alpha} dy = \frac{\alpha}{\alpha-1},$$

whereas

$$\mathbb{E}[Y_t^2] = \alpha \int_1^\infty y^{1-\alpha} dy = \infty$$

because $\alpha < 2$. Hence $\mathbb{E}[\varepsilon_t] = 0$, and

$$\mathbb{E}|\varepsilon_t| \leq \mathbb{E}[Y_t] + m < \infty.$$

Moreover, if $Y_t \geq 2m$, then $(Y_t - m)^2 \geq Y_t^2/4$, so

$$\mathbb{E}[\varepsilon_t^2] \geq \frac{1}{4} \mathbb{E}[Y_t^2 \mathbf{1}\{Y_t \geq 2m\}] = \infty.$$

2. Linearity and the triangle inequality give

$$\mathbb{E}[X_t] = 0, \quad \mathbb{E}|X_t| \leq (1 + |\rho|)\mathbb{E}|\varepsilon_0| < \infty.$$

To verify the second moment carefully, choose $K > 0$ such that $\mathbb{P}(|\rho\varepsilon_{t-1}| \leq K) > 0$. By independence,

$$\begin{aligned} \mathbb{E}[X_t^2] &\geq \mathbb{E}[(\varepsilon_t + \rho\varepsilon_{t-1})^2 \mathbf{1}\{|\rho\varepsilon_{t-1}| \leq K, |\varepsilon_t| > 2K\}] \\ &\geq \frac{1}{4} \mathbb{P}(|\rho\varepsilon_{t-1}| \leq K) \mathbb{E}[\varepsilon_t^2 \mathbf{1}\{|\varepsilon_t| > 2K\}] = \infty. \end{aligned}$$

3. X_t is measurable with respect to $\sigma(\varepsilon_t, \varepsilon_{t-1})$. If $|t - s| > 1$, the two corresponding pairs of innovations are disjoint and hence independent. Therefore X_t and X_s are independent.

4. Expanding and shifting indices,

$$\begin{aligned} \sum_{t=1}^n X_t &= \sum_{t=1}^n \varepsilon_t + \rho \sum_{t=1}^n \varepsilon_{t-1} \\ &= (1 + \rho) \sum_{t=1}^{n-1} \varepsilon_t + \varepsilon_n + \rho\varepsilon_0. \end{aligned}$$

5. Since the innovations are i.i.d., integrable, and mean zero, the SLLN gives

$$\frac{1}{n} \sum_{t=1}^{n-1} \varepsilon_t \longrightarrow 0 \quad \text{a.s.}$$

Writing $S_n = \sum_{t=1}^n \varepsilon_t$,

$$\frac{\varepsilon_n}{n} = \frac{S_n}{n} - \frac{n-1}{n} \frac{S_{n-1}}{n-1} \longrightarrow 0 \quad \text{a.s.}$$

Also $\rho\varepsilon_0/n \rightarrow 0$ a.s.

6. Dividing the identity in part 4 by (n) and applying part 5 yields

$$\boxed{\frac{1}{n} \sum_{t=1}^n X_t \longrightarrow 0 \quad \text{a.s.}}$$

Although X_t has infinite variance, the proof only needs integrability of the underlying i.i.d. innovations. A finite-variance version of the SLLN is therefore not directly applicable to X_t .

2 The Classical Coupon-Collector Problem

1. The time spent moving from (k-1) observed coupon types to (k) types is $W_{k,n} = \tau_k - \tau_{k-1}$. The increments telescope:

$$\sum_{k=1}^n W_{k,n} = \tau_n - \tau_0 = T_n.$$

2. After (k-1) distinct types have appeared, (n-k+1) types remain. Thus

$$p_{k,n} = \frac{n-k+1}{n}, \quad W_{k,n} \sim \text{Geom}(p_{k,n}).$$

3. By independence and the mean of a geometric variable,

$$\mathbb{E}[T_n] = \sum_{k=1}^n \frac{1}{p_{k,n}} = n \sum_{j=1}^n \frac{1}{j} = nH_n.$$

Since $H_n/\log n \rightarrow 1$,

$$\frac{\mathbb{E}[T_n]}{n \log n} \longrightarrow 1.$$

4. With $j = n - k + 1$,

$$\begin{aligned} \text{Var}(T_n) &= \sum_{k=1}^n \frac{1 - p_{k,n}}{p_{k,n}^2} \\ &= \sum_{j=1}^n \frac{1 - j/n}{(j/n)^2} \leq n^2 \sum_{j=1}^n \frac{1}{j^2} \leq \frac{\pi^2}{6} n^2. \end{aligned}$$

5. Consequently,

$$\text{Var} \left(\frac{T_n}{n \log n} \right) \leq \frac{\pi^2}{6 \log^2 n} \rightarrow 0.$$

For every $\varepsilon > 0$, Chebyshev's inequality gives

$$\mathbb{P} \left(\left| \frac{T_n - \mathbb{E}[T_n]}{n \log n} \right| > \varepsilon \right) \leq \frac{\pi^2}{6\varepsilon^2 \log^2 n} \rightarrow 0.$$

6. Decompose

$$\frac{T_n}{n \log n} = \frac{T_n - \mathbb{E}[T_n]}{n \log n} + \frac{\mathbb{E}[T_n]}{n \log n}.$$

The first term converges to zero in probability and the second to one. Therefore

$$\boxed{\frac{T_n}{n \log n} \rightarrow 1 \text{ in probability.}}$$

Thus the collection time is asymptotically $n \log n$ on its natural first-order scale.

3 A Two-Series Theorem for Positive Random Variables

Fix $c > 0$ and put $W_n = V_n \mathbf{1}\{V_n \leq c\}$. Kolmogorov's three-series theorem says that $\sum_n V_n$ converges a.s. if and only if

$$\sum_n \mathbb{P}(V_n > c) < \infty, \quad \sum_n \mathbb{E}[W_n] \text{ converges,} \quad \sum_n \text{Var}(W_n) < \infty.$$

Because $W_n \geq 0$, convergence of the second series is exactly $\sum_n \mathbb{E}[W_n] < \infty$. Furthermore,

$$\text{Var}(W_n) \leq \mathbb{E}[W_n^2] \leq c \mathbb{E}[W_n].$$

Hence the variance series follows automatically from the expectation series. The three-series criterion therefore reduces to

$$\boxed{\sum_n V_n < \infty \text{ a.s.} \iff \sum_n \mathbb{P}(V_n > c) < \infty \text{ and } \sum_n \mathbb{E}[V_n \mathbf{1}\{V_n \leq c\}] < \infty.}$$

4 Equivalent Criteria for Summability

For every $x \geq 0$,

$$\frac{1}{2} \min\{x, 1\} \leq \frac{x}{1+x} \leq \min\{x, 1\}.$$

Indeed, consider separately $0 \leq x \leq 1$ and $x > 1$. Taking expectations and summing shows immediately that

$$\sum_n \mathbb{E} \left[\frac{X_n}{1+X_n} \right] < \infty \iff \sum_n \mathbb{E}[\min\{X_n, 1\}] < \infty.$$

If the latter holds, Tonelli's theorem gives

$$\mathbb{E} \left[\sum_n \min\{X_n, 1\} \right] < \infty,$$

so $\sum_n \min\{X_n, 1\} < \infty$ a.s. In particular, $X_n > 1$ occurs only finitely often, and hence

$$\sum_n X_n < \infty \quad \text{a.s.}$$

Conversely, if $\sum_n X_n < \infty$ a.s., the two-series result with $c = 1$ yields

$$\sum_n \mathbb{E}[X_n \mathbf{1}\{X_n \leq 1\}] < \infty, \quad \sum_n \mathbb{P}(X_n > 1) < \infty.$$

Since

$$\mathbb{E}[\min\{X_n, 1\}] = \mathbb{E}[X_n \mathbf{1}\{X_n \leq 1\}] + \mathbb{P}(X_n > 1),$$

condition (iii) follows. Thus (i), (ii), and (iii) are equivalent.

5 The Strong Law and L^1 Convergence

Let $\mu = \mathbb{E}[X_1]$. Since $X_1 \in L^1$, the de la Vallée–Poussin criterion provides an increasing convex function $\Phi : [0, \infty) \rightarrow [0, \infty)$ such that

$$\frac{\Phi(t)}{t} \rightarrow \infty, \quad \mathbb{E}\Phi(|X_1|) < \infty.$$

By convexity and Jensen's inequality,

$$\Phi\left(\left|\frac{S_n}{n}\right|\right) \leq \Phi\left(\frac{1}{n} \sum_{i=1}^n |X_i|\right) \leq \frac{1}{n} \sum_{i=1}^n \Phi(|X_i|).$$

Therefore

$$\sup_n \mathbb{E}\Phi\left(\left|\frac{S_n}{n}\right|\right) \leq \mathbb{E}\Phi(|X_1|) < \infty.$$

Another application of the de la Vallée–Poussin criterion shows that $\{S_n/n : n \geq 1\}$ is uniformly integrable. The SLLN gives $S_n/n \rightarrow \mu$ a.s., hence in probability. Uniform integrability then upgrades this convergence to L^1 :

$$\boxed{\mathbb{E}\left|\frac{S_n}{n} - \mu\right| \rightarrow 0.}$$

6 The Empirical Distribution Function at a Fixed Point

1. Put $I_i(x) = \mathbf{1}\{X_i \leq x\}$. Then the $I_i(x)$ are i.i.d. Bernoulli($F(x)$), so

$$\mathbb{E}[\hat{F}_n(x)] = F(x), \quad \text{Var}(\hat{F}_n(x)) = \frac{F(x)(1 - F(x))}{n}.$$

2. For each fixed x , the SLLN applied to $I_i(x)$ gives

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I_i(x) \rightarrow F(x) \quad \text{a.s.}$$

3. Independence across sample indices implies

$$\text{Cov}(\widehat{F}_n(x), \widehat{F}_n(y)) = \frac{1}{n} \text{Cov}(I_1(x), I_1(y)).$$

Since $I_1(x)I_1(y) = I_1(x \wedge y)$,

$$\boxed{\text{Cov}(\widehat{F}_n(x), \widehat{F}_n(y)) = \frac{F(x \wedge y) - F(x)F(y)}{n}.}$$

4. Pointwise convergence controls each predetermined x , while the supremum ranges over an uncountable set and its maximizing point may depend on the sample and on n . Thus pointwise convergence alone does not imply uniform convergence; an additional argument exploiting monotonicity or empirical-process bounds is required.

7 A Uniform Law over a Finite Class

1. For fixed f , the variables $f(X_i)$ are independent and lie in $[0, 1]$. Hoeffding's inequality gives

$$\mathbb{P}(|P_n f - P f| > \varepsilon) \leq 2e^{-2n\varepsilon^2}.$$

2. By the union bound,

$$\begin{aligned} \mathbb{P}\left(\sup_{f \in \mathcal{F}} |P_n f - P f| > \varepsilon\right) &\leq \sum_{j=1}^M \mathbb{P}(|P_n f_j - P f_j| > \varepsilon) \\ &\leq 2Me^{-2n\varepsilon^2}. \end{aligned}$$

3. Set the last display equal to δ . Solving for ε gives, with probability at least $1 - \delta$,

$$\boxed{\sup_{f \in \mathcal{F}} |P_n f - P f| \leq \sqrt{\frac{\log(2M/\delta)}{2n}}.}$$

4. For each fixed $\varepsilon > 0$,

$$\mathbb{P}\left(\sup_{f \in \mathcal{F}_n} |P_n f - P f| > \varepsilon\right) \leq 2 \exp\{\log M_n - 2n\varepsilon^2\}.$$

This tends to zero whenever

$$\boxed{\log M_n = o(n).}$$

Thus this condition guarantees convergence in probability.

8 Rademacher Complexity and Uniform Convergence

Write $P'_n f = n^{-1} \sum_{i=1}^n f(X'_i)$.

1. Since $Pf = \mathbb{E}[P'_n f \mid X_1, \dots, X_n]$, conditional Jensen's inequality gives

$$\sup_{f \in \mathcal{F}} |P_n f - Pf| \leq \mathbb{E} \left[\sup_{f \in \mathcal{F}} |P_n f - P'_n f| \mid X_1, \dots, X_n \right].$$

Taking expectation proves the claim.

2. Put $\mathcal{F}_\pm = \mathcal{F} \cup (-\mathcal{F})$. Then

$$\sup_{f \in \mathcal{F}} |(P_n - P'_n)f| = \sup_{f \in \mathcal{F}_\pm} (P_n - P'_n)f.$$

The joint distribution is unchanged when each pair (X_i, X'_i) is swapped according to an independent Rademacher sign. Hence

$$\begin{aligned} \mathbb{E} \sup_{f \in \mathcal{F}_\pm} (P_n - P'_n)f &= \mathbb{E} \sup_{f \in \mathcal{F}_\pm} \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(X_i) - f(X'_i)) \\ &\leq \mathbb{E} \sup_{f \in \mathcal{F}_\pm} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) + \mathbb{E} \sup_{f \in \mathcal{F}_\pm} \frac{1}{n} \sum_{i=1}^n (-\varepsilon_i) f(X'_i) \\ &= 2\mathbb{E}[\hat{\mathfrak{R}}_n(\mathcal{F}_\pm)]. \end{aligned}$$

3. Conditional on the data, independence of the signs and $\cosh u \leq e^{u^2/2}$ give

$$\begin{aligned} \mathbb{E}_\varepsilon \exp \left\{ \lambda \sum_{i=1}^n \varepsilon_i f_j(X_i) \right\} &= \prod_{i=1}^n \cosh(\lambda f_j(X_i)) \\ &\leq \exp \left\{ \frac{\lambda^2}{2} \sum_{i=1}^n f_j(X_i)^2 \right\}. \end{aligned}$$

4. Let $R = \max_j (\sum_i f_j(X_i)^2)^{1/2}$. For every $\lambda > 0$,

$$\begin{aligned} \mathbb{E}_\varepsilon \max_{1 \leq j \leq M} \sum_{i=1}^n \varepsilon_i f_j(X_i) &\leq \frac{1}{\lambda} \log \sum_{j=1}^M \mathbb{E}_\varepsilon e^{\lambda \sum_i \varepsilon_i f_j(X_i)} \\ &\leq \frac{\log M}{\lambda} + \frac{\lambda R^2}{2}. \end{aligned}$$

Optimizing at $\lambda = \sqrt{2 \log M}/R$ and dividing by n yields

$$\hat{\mathfrak{R}}_n(\mathcal{F}) \leq \frac{R \sqrt{2 \log M}}{n}.$$

Since $0 \leq f_j \leq 1$, $R \leq \sqrt{n}$, and therefore

$$\hat{\mathfrak{R}}_n(\mathcal{F}) \leq \sqrt{\frac{2 \log M}{n}}.$$

5. The class $\mathcal{F}_\pm$ has at most $2M$ elements and all its functions have absolute value at most one. Parts 2 and 4 give

$$\boxed{\mathbb{E} \sup_{f \in \mathcal{F}} |P_n f - Pf| \leq 2 \sqrt{\frac{2 \log(2M)}{n}}}.$$

6. Define

$$Z(X_1, \dots, X_n) = \sup_{f \in \mathcal{F}} |P_n f - P f|.$$

Replacing one observation changes every $P_n f$ by at most $1/n$, and hence changes (Z) by at most $(1/n)$. McDiarmid's inequality gives

$$\mathbb{P}(Z - \mathbb{E}Z > t) \leq e^{-2nt^2}.$$

Taking $t = \sqrt{\log(1/\delta)/(2n)}$ and using part 5 proves that, with probability at least $1 - \delta$,

$$Z \leq 2\sqrt{\frac{2 \log(2M)}{n}} + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

Rademacher complexity measures how well a class can correlate with random signs on the observed sample. A richer class can fit these signs more closely and has larger complexity; a smaller complexity therefore yields a tighter uniform deviation bound.